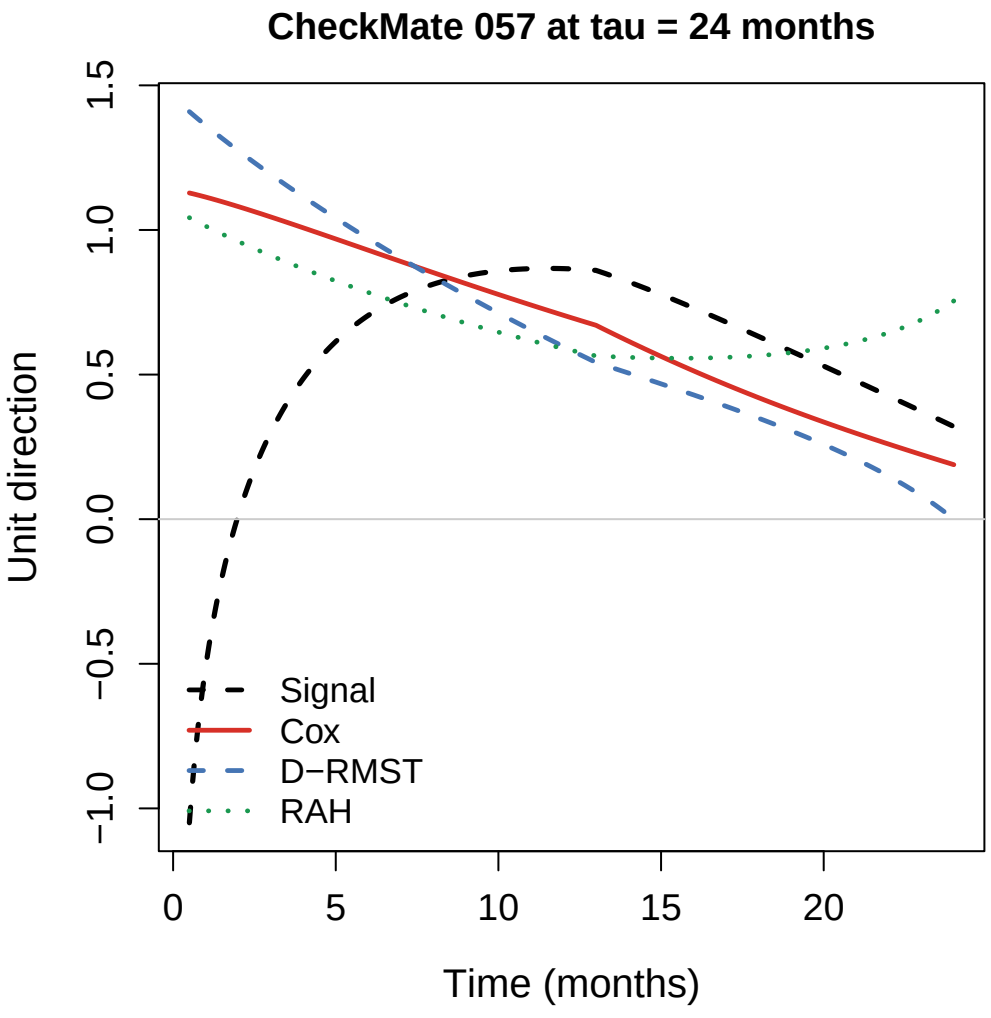
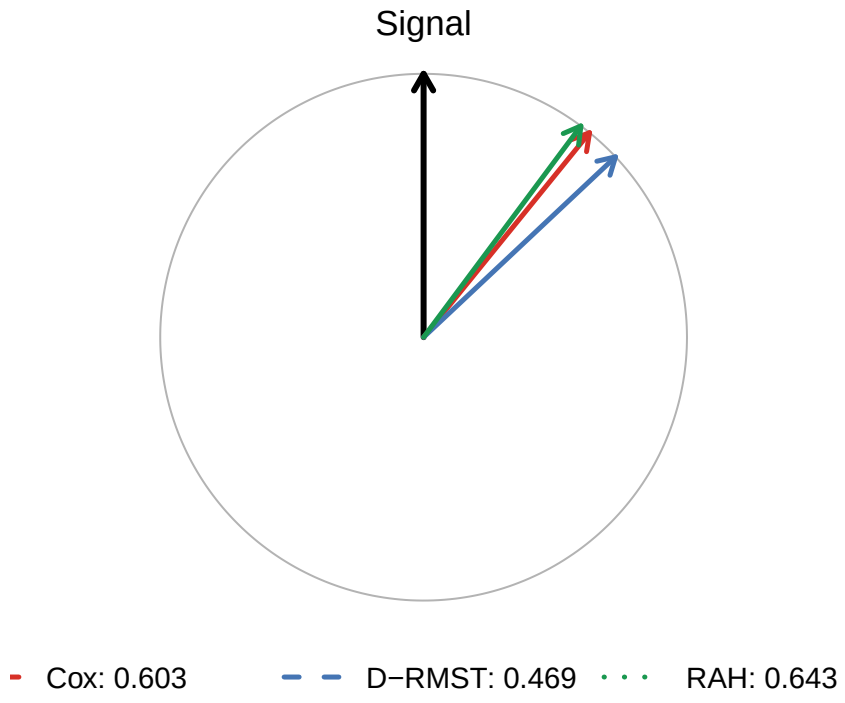
$$\cos^{2(\theta_X)} = \xi_X / |s_\alpha|^2$$